

Write a program for the task of Credit Score Classification

CODE:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

data = {
    'Income': [25000, 30000, 45000, 50000, 60000, 70000,
               28000, 35000, 55000, 80000, 40000, 65000],
    'Age': [22, 25, 30, 35, 40, 45, 24, 28, 38, 50, 32, 42],
    'Credit_History': [1, 1, 0, 1, 1, 1,
                       0, 1, 1, 1, 0, 1],
    'Credit_Score': [0, 0, 0, 1, 1, 1,
                     0, 0, 1, 1, 0, 1]
}

df = pd.DataFrame(data)
X = df[['Income', 'Age', 'Credit_History']]
y = df['Credit_Score']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Credit Score Classification")
print("-----")
```

```
print("\nActual Values:")
print(y_test.values)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy:", accuracy)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
new_customer = [[55000, 35, 1]]
prediction = model.predict(new_customer)

if prediction[0] == 1:
    print("\nNew Customer Credit Score: GOOD")
else:
    print("\nNew Customer Credit Score: POOR")
```

OUTPUT:

... Credit Score Classification

Actual Values:

[0 1 0]

Predicted Values:

[0 1 0]

Accuracy: 1.0

Confusion Matrix:

[[2 0]

[0 1]]

New Customer Credit Score: GOOD